MATTHEW AZER

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EDUCATION

Rutgers University - School of Engineering **New Brunswick, NJ**
Bachelor of Science in Electrical & Computer Engineering ***Expected May 2028***

Relevant Coursework
Integrated Data-Driven Design for Engineering Applications, *Engineering Mechanics: Statics*, *Principles of Electrical Engineering*, *Digital Logic Design*, *Probability & Random Processes*, *Programming Methodology*, *Computer Architecture*, *Linear Systems & Signals*

WORK EXPERIENCE

Breeze-Eastern LLC **Whippany, NJ**
Manufacturing Engineer Intern ***Nov 2026 - Present***

- Developing a Python-based automation system for laser-marked aerospace data plates, automatically generating, validating, and tracking unique serial numbers across 100+ hoist and cargo-hook assemblies, targeting a 50%+ reduction in manual processing time and improved production traceability.
- Support 5+ engineers with electrical testing and troubleshooting of aerospace hoist and cargo-hook systems, using multimeters, serial interfaces, schematics, and diagnostic equipment to identify and resolve 20+ hardware, wiring, and integration issues.
- Perform Linux-based system integration and diagnostics across electrical hardware, test equipment, and production software, using Bash/Python scripting, serial communications, log analysis, process monitoring, and interface troubleshooting to isolate failures across 10+ hardware/software configurations.

MasTec Utility Services (PSE&G Contract) **New Brunswick, NJ**
Field Technician Intern ***May 2026 - Aug 2026***

- Performed cathodic protection testing across 500+ test stations, collecting and analyzing 500+ voltage measurements to assess corrosion protection and identify 25+ abnormal conditions on PSE&G underground gas pipeline systems.
- Utilized digital reporting software to document and validate 500+ field measurements, verify measurement accuracy, and assist in troubleshooting abnormal system readings, reducing repeat field measurements by approximately 15%.

PROJECTS

Autonomous Word Hunt Robot

- Developed a Python-based computer vision system using OpenCV and Tesseract OCR to recognize a live 4x4 letter grid and identify high-scoring words from a database of 129,000+ candidate words using trie-based depth-first search and path-planning algorithms.
- Programmed a GRBL-controlled XY plotter with a servo-actuated capacitive stylus to autonomously trace and submit 45+ words per game, optimizing motion speed, acceleration, and touchscreen interaction for reliable operation.

Four-Way Traffic Light Controller

- Designed and built a four-way traffic light control circuit using AND/OR/NOT gates, JK/D flip-flops, counters, and a 555 timer to automatically sequence 12 red, yellow, and green LEDs across an intersection.
- Implemented and tested sequential and combinational logic on a breadboard, troubleshooting circuit timing, wiring, and logic errors to achieve reliable operation across 100+ continuous signal cycles.

LEADERSHIP & INVOLVEMENT

N2E Robotics Club (Rutgers University) **New Brunswick, NJ**
Member ***Sep 2024 - Present***

- Participate in hands-on robotics projects involving ROS, simulation, sensors, and programming, collaborating with students to build, configure, test, and troubleshoot autonomous robotic systems.
- Collaborate with other students during technical workshops to troubleshoot hardware and software, configure robotic systems, and develop solutions for autonomous and simulated robot applications.

SKILLS

Languages: *Python*, *C*, *C++*, *Java*, *MATLAB*, *Verilog*, *RISC-V Assembly*, *G-code*

Hardware & EDA Tools: *LTspice*, *Xilinx Vivado*, *Arduino IDE*, *Raspberry Pi*, *AutoCAD*, *SolidWorks*, *Onshape*, *IBM Rhapsody*, *Soldering*, *Circuit Testing*, *Digital Multimeter*, *Frequency Measurement*, *Serial Communication*

Software & Libraries: *Linux*, *OpenCV*, *Tesseract OCR*, *Git/GitHub*, *VS Code*, *Cursor*, *ROS*